

10 CFR 50.73

RA18-024

April 16, 2018

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

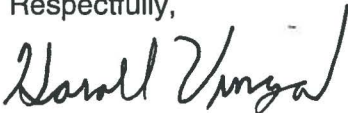
LaSalle County Station, Unit 1 and Unit 2
Renewed Facility Operating Licenses No. NPF-11 and NPF-18
NRC Docket No. 50-373 and No. 50-374

Subject: Licensee Event Report 2018-001-00, Unanalyzed Condition Affecting
Accident Mitigation for Tornado Generated Missile Protection Non-
Conformances

In accordance with 10 CFR 50.73(a)(2)(i)(B), 10 CFR 50.73(a)(2)(ii)(B), 10 CFR 50.73(a)(2)(v)(D), 10 CFR 50.73(a)(2)(vii), and 10 CFR 50.73(a)(2)(ix)(A), Exelon Generation Company, LLC (EGC) is submitting Licensee Event Report (LER) Number 2018-001-00 for LaSalle County Station, Unit 1 and Unit 2.

There are no regulatory commitments in this letter. Should you have any questions concerning this report, please contact Mr. Guy V. Ford, Jr., Regulatory Assurance Manager, at (815) 415-2800.

Respectfully,



Harold T. Vinyard
Plant Manager
LaSalle County Station

Enclosure: Licensee Event Report

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollect.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NE08-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

LaSalle County Station, Unit 1

2. Docket Number

05000373

3. Page

1 OF 5

4. Title

Unanalyzed Condition Affecting Accident Mitigation for Tornado Generated Missile Protection Non-Conformances

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
02	15	18	2018	- 001	- 00	04	16	18	LaSalle County Station, Unit 2	05000374
									Facility Name	Docket Number
									NA	NA
9. Operating Mode			11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)							
1			☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)				
			☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☒ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)				
			☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☒ 50.73(a)(2)(ix)(A)				
			☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)				
10. Power Level			☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)				
95			☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)				
			☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)				
			☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)				
			☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(iii)				
			☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)						

12. Licensee Contact for this LER**Licensee Contact**

John Keenan, Engineering Director

Telephone Number (Include Area Code)

(815) 415-3800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
NA	NA	NA	NA	Yes	NA	NA	NA	NA	NA

14. Supplemental Report Expected☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No**15. Expected Submission Date**

Month	Day	Year
NA	NA	NA

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On February 15, 2018, during evaluation of protection for Technical Specifications (TS) equipment from the damaging effects of tornado generated missiles, LaSalle County Station (LSCS) identified non-conforming conditions in the plant design such that specific TS equipment is considered inadequately protected from tornado generated missiles. Specifically, tornado generated missiles could strike the components supporting the operation of the main control room ventilation (VC) and auxiliary electric equipment room ventilation (VE) systems, and the Unit 2 Division 2 480-volt motor control center 236X-1 (2AP78E). The VC/VE systems provide a protected environment for occupants to control the unit following an uncontrolled release of radioactivity, hazardous chemicals, or smoke if a tornado were to occur. This condition is a legacy design issue that has existed since the original licensing of the plant.

This condition is reportable under multiple reporting criteria in accordance with 10CFR50.73 requirements indicated above. This condition is being addressed in accordance with NRC enforcement guidance provided in Enforcement Guidance Memorandum (EGM) 15-002, Revision 1, "Enforcement Discretion for Tornado-Generated Missile Protection Noncompliance," and DSS-ISG-2016-01, Revision 1, "Clarification of Licensee Actions in Receipt of Enforcement Discretion per Enforcement Guidance Memorandum EGM 15-002." Compensatory measures have been implemented in accordance with these documents.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 1	05000373	2018	- 001	- 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

LaSalle County Station (LSCS) Unit 1 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power.

The affected systems are common (both units) ventilation required for the main control room (MCR) and auxiliary electric equipment room (AEER). These systems provide necessary ventilation and a protected environment for occupants to control the unit(s) following an uncontrolled release of radioactivity, hazardous chemicals, or smoke if a tornado were to occur. In addition, the Unit 2 division 2 480V power supply motor control center (MCC) 236X-1 was affected, which impacted various loads on Unit 2 including Unit 2 standby gas treatment, Unit 2 post-LOCA, "B" MCR area filtration system supply and exhaust fan, reactor building division 2 isolation dampers control logic, Unit 2 battery room exhaust fan, and Unit 2 24/48VDC battery rooms exhaust fans.

CONDITION PRIOR TO EVENT

Unit(s): 1 / 2 Date: February 15, 2018 Time: 1100 CST
Reactor Mode(s): 1 / 1 Mode(s) Name: Power Operation / Power Operation Power Level: 95 percent / 100 percent

DESCRIPTION

On February 15, 2018, during evaluation of protection for Technical Specifications (TS) equipment from the damaging effects of tornado generated missiles, LSCS identified non-conforming conditions in the plant design such that specific TS equipment is considered inadequately protected from tornado generated missiles. Specifically, tornado generated missiles could strike the components supporting the operation of MCR ventilation (VC) and AEER ventilation (VE) systems. This could result in inoperable VC/VE systems, which provide a protected environment for occupants to control the unit(s) following an uncontrolled release of radioactivity, hazardous chemicals, or smoke if a tornado were to occur.

Station personnel entered this condition into the corrective action program and notified the NRC via Event Notification 53213 on February 15, 2018. There were no structures, systems, or components (SSCs) out of service that contributed to this condition. The specific SSCs found to be non-conforming relative to tornado-missile design basis requirements are:

- Unit 1(2) VC "A" train (19 components)
- Unit 1(2) VC "B" train (19 components)
- Unit 1(2) VE "A" train (12 components)
- Unit 1(2) VE "B" train (12 components)
- Division 2 480V MCC 236X-1 (2AP78E)

CAUSE

The cause of the non-conforming conditions is a legacy design issue that was identified as part of response to Regulatory Issue Summary (RIS) 2015-06, "Tornado Missile Protection," and Enforcement Guidance Memorandum (EGM) 15-002, "Enforcement Discretion for Tornado Generated Missile Protection Noncompliance." These conditions were part of the original plant design and have existed since the initial construction and licensing of the plant. LSCS received its operating licenses for Unit 1 (NPF-11) and Unit 2 (NPF-18) on April 17, 1982. These operating licenses were issued based, in part, on the design as described in the Final Safety Analysis Report (FSAR).

REPORTABILITY AND SAFETY ANALYSIS

This condition was previously reported in accordance with 10 CFR 50.72(b)(3)(ii)(B) as a condition that results in the nuclear power plant being in an unanalyzed condition that significantly degrades plant safety and 10 CFR 50.72(b)(3)(v)(D) as a condition that at the time of discovery could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident. This condition was reported to the NRC via Event Notification System (ENS) report number 53213 on February 15, 2018. There were no actual consequences resulting from these identified non-conforming conditions.

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NARRATIVE

This condition is reportable as a licensee event report (LER) in accordance with:

- 10CFR 50.73(a)(2)(i)(B) for operation or condition prohibited by TS, since the design deficiency (i.e., non-conforming condition) existed for a time longer than permitted by TS;
- 10CFR 50.73(a)(2)(ii)(B) for an unanalyzed condition that significantly degraded plant safety;
- 10CFR 50.73(a)(2)(v)(D) for an event or condition that would have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident since the VC and VE system are part of the habitability environment for operators used in response to a postulated radioactive release;
- 10CFR 50.73(a)(2)(vii) for a single event that affected multiple trains in a single system used to mitigate the consequences of an accident since the design deficiency affected both trains of VC and both trains of VE; and,
- 10CFR 50.73(a)(2)(ix)(A) for an event or condition that affected two or more trains in different systems used to mitigate the consequences of an accident since the design deficiency affected both VC and VE systems, as well as Unit 2 Division 2 systems supported by 480V MCC 236X-1.

Current Licensing Basis (CLB) Review

The affected SSCs are required for Tornado Safe Shutdown (TSS) but are not protected against tornado missiles as required to meet the CLB. The LSCS CLB for tornados and tornado missiles pertinent to RIS 2015-06 are described in Updated Final Safety Analysis Report (UFSAR). The original plant licensing documents were reviewed to determine licensing requirements that are not specifically detailed in the LSCS UFSAR. The additional documents which were reviewed include the original Standard Review Plan (NUREG-0800), Safety Evaluation Report (NUREG-0519), FSAR, FSAR Questions and Answers applicable to tornados and tornado missile protection, and the General Design Criteria (GDC).

The design basis tornado has a maximum rotational velocity of 300 miles per hour (mph), a translational velocity of 60 mph, causes a pressure drop of 3 pounds per square inch (psi) at the vortex in three seconds, and a radius of maximum wind speed of 227 feet. The CLB for tornado missile protection design involves two types of missiles which have been considered. They are a wood plank of 4 inches by 12 inches by 12 feet dimension with an impact velocity of 225 mph; and a 4,000-pound automobile with a 20-square foot front area and a 50-mph impact velocity. Station personnel reviewed these requirements against the plant SSCs needed to achieve safe shutdown in the event of a tornado. The result was the identification of non-conforming conditions described above in the event description.

Operability Discussion

Due to the identified vulnerabilities from a tornado missile, both trains of the MCR and the AEER Control Room Area Filtration systems, the Control Room Area Ventilation Air Condition Systems, and Unit 2 MCC 236X-1 were declared inoperable. For both Unit 1 and Unit 2, the station entered TS 3.7.4 Required Actions A.1 and E.1 and TS 3.7.5 Required Actions A.1 and B.1. Although the probability of loss of function is low, it is possible that tornado missiles could impact the MCR and/or the AEER ventilation systems, resulting in loss of the TS 3.7.4 and or 3.7.5 safety function for both trains of VC and/or VE.

Division 2 480V MCC 236X-1 being inoperable resulted in the following equipment fed from the MCC also being INOPERABLE which includes U2 SBGT, U2 Division 2 Post LOCA, B MCR Control Room Area Filtration system supply and exhaust fan, Reactor Building Division 2 isolation Dampers 2VR04YB and 2VR05YA control logic, 2VQ037 damper, Unit 2 Division 2 battery room exhaust fan and Unit 2 24/48 Volt battery rooms exhaust fans. The station entered TS 3.8.7 Required Action A.1 for Loss of 236X-1 Power Distribution Subsystem for Unit 2, TS 3.8.7 Required Action D.1 for Loss of 236X-1 Power Distribution Subsystem for Unit 1, TS 3.7.4 Required Action A.1 for the B MCR and the AEER Control Room Area Filtration systems, previously entered for both units, TS 3.6.4.3 Required Action A.1 for Unit 2 SBGT for both units, and TS 3.6.4.2 Required Actions A.1 and A.2 for 2VQ037 and Dampers 2VR04YB and 2VR05YA both units. Inoperability of battery room fans for U-2 Division 2 and 24/48 Volt batteries is addressed under guidance in LOP-VX-02 without affecting operability of the associated DC systems.

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NARRATIVE

Based on the tornado analysis, the MCR and AEER ventilation systems and U-2 Division 2 480V MCC 236X-1 and respective equipment listed above were inoperable, with the applicable TSs immediately exited per allowance of EGM 15-002, Revision 1, that states the listed tornado vulnerable equipment is considered operable, but non-conforming. LSCS exited TS 3.7.4 Required Actions and TS 3.7.5 Required Actions for the MCR and the AEER Control Room Area Filtration systems and the Control Room Area Ventilation Air Condition systems, TS 3.8.7 Electrical Distribution Systems, TS 3.6.4.3 Standby Gas Treatment System, and TS 3.6.4.2 Secondary Containment Isolation Valves. The battery room fans for U-2 Division 2 and 24/48 volt batteries were also considered operable but non-conforming. This EGM applies to a SSC that is determined to be inoperable for tornado generated missile protection. In addition, Interim Staff Guidance DSS-ISG-2016-01, Revision 1, "Clarification of Licensee Actions in Receipt of Enforcement Discretion per Enforcement Guidance Memorandum EGM 15-002," provides clarification for requirements to be met to allow enforcement discretion. In accordance with Appendix A, Clarification of Actions in EGM 15-002, Revision 1, initial compensatory actions have been established that provide additional protection such that the tornado missile effects are lessened.

Application of EGM 15-002

Application of the guidance in EGM 15-002, Revision 1, and, NRC DSS-ISG-2016-01, Revision 1, Appendix A, describes acceptable initial and comprehensive compensatory measures necessary to ensure that the identified nonconforming conditions can be treated by the NRC with enforcement discretion. The non-conforming conditions will require resolution prior to the end of the enforcement discretion period, which is June 10, 2018 since LSCS is classified as a "Group A" (Higher Tornado Missile Risk) plant. LSCS applied the guidance in EGM 15-002, Revision 1, and NRC DSS-ISG-2016-01, Revision 1, Appendix A, to determine the acceptable initial and comprehensive compensatory measures necessary to ensure that the identified non-conforming conditions could be treated by the NRC with enforcement discretion.

Subsequently, in accordance with the EGM 15-002, Revision 1, a request to extend the expiration date for the period of enforcement discretion for LSCS from June 10, 2018 to June 10, 2020 was submitted to the NRC on March 20, 2018. The requested enforcement discretion expiration date of June 10, 2020 would allow sufficient time to resolve the tornado missile protection non-conformances and restore the site to compliance. Resolution of the non-conforming conditions is expected to involve a combination of analysis and potentially plant modification installations.

CORRECTIVE ACTIONS

This condition is being addressed in accordance with NRC enforcement guidance provided in EGM 15-002, Revision 1 and DSS-ISG-2016-01, Revision 1. Enforcement discretion provided by EGM 15-002, Revision 1 remains in effect until June 10, 2018, for LSCS, at which time modifications, license amendments, or other actions must be implemented to fully resolve the issue. Compensatory measures have been implemented in accordance with these documents.

Initial compensatory measures were established by an Operations Standing Order in accordance with NRC DSS-ISG-2016-0 Revision 1, Appendix A regarding:

1. Procedures were verified to be put in place, with associated current training, for performing actions in response to a tornado.
2. Procedures were verified to be put in place, with associated current training, for actions to be taken if a tornado watch is issued for the area.
3. Procedures were verified to be put in place, with associated current training, for actions to be taken if a tornado warning is issued for the area.
4. Established a heightened level of station awareness and preparedness relative to identified tornado missile vulnerabilities.

The comprehensive compensatory measures were established by incorporating the Standing Order actions and adding additional detail to Operations procedure LOA-TORN-001, Revision 22, "High Winds / Tornado," for completing additional inspections and restoration actions on equipment vulnerable to tornado missile damage. The procedure directs operators to inspect VC/VE and 236X-1 for damage after high wind conditions are clear. If VC/VE and/or 236X-1 experience damage during the high wind event, operators are directed to perform additional response actions delineated in the LOA-TORN-001, Revision 22 procedure's attachments. These actions are not time-critical actions and can easily be performed in conjunction with other actions.

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NARRATIVE

Additionally, in accordance with the EGM 15-002, Revision 1, a request to extend the expiration date for the period of enforcement discretion for LSCS to June 10, 2020 was submitted to the NRC on March 20, 2018. The requested enforcement discretion expiration date would allow time to resolve the tornado missile protection non-conformances and restore the site to compliance.

PREVIOUS OCCURRENCES

There have been no similar events or conditions at the station in the previous three years.

COMPONENT FAILURE DATA

Manufacturer: NA
Device: NA
Component ID: Model NA